Package ‘MINTer’

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Title Malaria INTervention Emulator in R

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Description A comprehensive package for malaria intervention simulation and neural network-based emulation. Combines malariasimulation with trained GRU/LSTM models for rapid prediction of intervention outcomes.

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Contents

`MINTer-package` *MINTer: Malaria INTervention Emulator in R*

Description

A comprehensive package for malaria intervention simulation and neural network-based emulation. Combines malariasimulation with trained GRU/LSTM models for rapid prediction of intervention outcomes.

Author(s)

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`available_net_types` *Return the vector of net types contained in an ITN-parameter file*

Description

Return the vector of net types contained in an ITN-parameter file

Usage

```
available_net_types(itn_params_path = NULL)
```

Arguments

`itn_params_path`
: Path to an RDS created by MINTer

`calculate_overall_dn0` *Calculate overall dn0 in a single call*

Description

Supply the net-type usage mix as named arguments and get

- **dn0** - the weighted-average dn0 at the requested resistance level, and
- **itn_use** - the total usage proportion of pyrethroid-based ITNs.

Usage

```
calculate_overall_dn0(
  resistance_level,
  ...,
  itn_params_path = NULL,
  strict = FALSE
)
```

Arguments

`resistance_level`
: Numeric resistance level.

`...`
: Named numeric values giving the usage proportion for each net type, e.g. `pyrethroid_only = 0.4`.

`itn_params_path`
: Optional RDS file with ITN parameters (defaults to the bundled example).

`strict`
: Forwarded to `define_bednet_types()`. If `TRUE`, execution stops when any requested net type is missing from the parameter file (even after the initial name check).

`convert_db_results_to_dataframe` *Convert Database Results to Standard Dataframe*

Description

Convert Database Results to Standard Dataframe

Usage

```
convert_db_results_to_dataframe(raw_results, predictor, model_types)
```

Arguments

`raw_results`	Results from run_emulator_db
`predictor`	"prevalence" or "cases"
`model_types`	Vector of model types used

Value

Data frame with standardized format

`create_malariasim_scenarios` *Create scenarios for Malaria Simulation*

Description

Create scenarios for Malaria Simulation

Usage

```
create_malariasim_scenarios(
  eir,
  dn0_use,
  dn0_future,
  Q0,
  phi_bednets,
  seasonal,
  routine,
  itn_use,
  irs_use,
  itn_future,
  irs_future,
  lsm
)
```

Arguments

`eir`	Entomological Inoculation Rate vector
`dn0_use`	Current bednet effectiveness vector
`dn0_future`	Future bednet effectiveness vector
`Q0`	Human blood index vector
`phi_bednets`	Proportion of bites on humans in bed vector
`seasonal`	Seasonal transmission indicator (0/1) vector
`routine`	Routine distribution indicator (0/1) vector
`itn_use`	Current ITN usage vector
`irs_use`	Current IRS coverage vector
`itn_future`	Future ITN usage vector
`irs_future`	Future IRS coverage vector
`lsm`	Larval source management coverage vector

Value

A data frame of scenarios

`create_scenarios` *Create Scenarios for Emulator*

Description

Create Scenarios for Emulator

Usage

```
create_scenarios(...)
```

Arguments

`...`	Named parameters with values

Value

Data frame of scenarios

`create_scenario_plots` *Create Scenario Plots from Emulator Results*

Description

Creates plots for each scenario from emulator results dataframe.

Usage

```
create_scenario_plots(results, output_dir = NULL, plot_tight = FALSE)
```

Arguments

`results`	Data frame from run_malaria_emulator
`output_dir`	Directory to save plots (optional)
`plot_tight`	Use tight y-axis scaling (default FALSE)

Value

List of ggplot objects

create_unified_plot *Create Unified Plot for Emulator Results*

Description

Create Unified Plot for Emulator Results

Usage

```
create_unified_plot(
  plot_data,
  predictor,
  title_text,
  window_size = 14,
  plot_tight = FALSE
)
```

Arguments

plot_data	Data frame with timestep, value, type columns
predictor	"prevalence" or "cases"
title_text	Title for the plot
window_size	Window size for time scaling
plot_tight	Use tight y-axis scaling

Value

ggplot object

define_bednet_types *Define bednet types and create spline fits*

Description

Define bednet types and create spline fits

Usage

```
define_bednet_types(net_types, itn_params_path = NULL, strict = FALSE)
```

Arguments

net_types	Character vector of net types to process
itn_params_path	Path to RDS file containing ITN parameters. If NULL, uses bundled data.
strict	Allows for future expansion of dataset (defaul FALSE to use only available nets in RDS)

Value

Named list of smooth.spline objects, one for each net type `fetch_rolling_data` *Fetch Rolling Data from DuckDB*

Description

Fetch Rolling Data from DuckDB

Usage

```
fetch_rolling_data(
  db_path,
  table_name = "simulation_results",
  window_size = 14,
  param_index = NULL,
  predictor = "prevalence"
)
```

Arguments

`db_path`	Path to DuckDB database
`table_name`	Name of table in database
`window_size`	Window size for rolling average (days)
`param_index`	Parameter index to fetch (NULL for list of available)
`predictor`	"prevalence" or "cases"

Value

Data frame with results or list of available parameters

`generate_scenario_predictions` *Generate Scenario Predictions*

Description

Generate Scenario Predictions

Usage

```
generate_scenario_predictions(
  scenarios,
  models,
  model_types = c("GRU", "LSTM"),
  time_steps = 2190
)
```

Arguments

`scenarios`	Data frame with scenario parameters
`models`	List from load_emulator_models()
`model_types`	Vector of model types to use ("GRU", "LSTM", or both)
`time_steps`	Number of time steps to predict (in days for prevalence)

Value

List with predictions for each scenario

---

get_runtime_parameters *Get Runtime Parameters for Simulation*

Description

Get Runtime Parameters for Simulation

Usage

```
get_runtime_parameters(
  i,
  lhs_data,
  HUMAN_POPULATION,
  bednet_params,
  SIM_LENGTH
)
```

Arguments

`i`	Index of parameter set
`lhs_data`	Data frame of LHS scenarios
`HUMAN_POPULATION`	Human population size
`bednet_params`	Bednet parameter data
`SIM_LENGTH`	Simulation length in days

Value

List of input parameters

initialize_python *Initialize Python Dependencies*

Description

This function initialises the Python dependencies required for the emulator. It is called automatically when needed, but can be called manually to pre-load the dependencies.

Usage

```
initialize_python(verbose = TRUE)
```

Arguments

verbose	Logical, whether to print messages (default: TRUE)

initialize_simulation_parameters *Initialize Simulation Parameters*

Description

Initialize Simulation Parameters

Usage

```
initialize_simulation_parameters(
  lhs_sample,
  HUMAN_POPULATION,
  selected_seasonality
)
```

Arguments

lhs_sample	Single row from LHS data
HUMAN_POPULATION	Human population size
selected_seasonality	Seasonality parameters

Value

malariasimulation parameters object

`list_available_parameters` *List Available Parameters in Database*

Description

List Available Parameters in Database

Usage

```
list_available_parameters(db_path, table_name = "simulation_results")
```

Arguments

`db_path`	Path to DuckDB database
`table_name`	Name of table in database

Value

Data frame with parameter_index and global_index

`load_emulator_models` *Load Emulator Models*

Description

Load Emulator Models

Usage

```
load_emulator_models(
  models_base_dir = NULL,
  predictor = "prevalence",
  device = NULL
)
```

Arguments

`models_base_dir`	Base directory containing model files. If NULL (default), uses models bundled with the package.
`predictor`	"prevalence" or "cases"
`device`	"cpu" or "cuda" (NULL for auto-detect)

Value

List containing models and configuration

local_cluster_malariasim_controller

Run Malaria Simulation with Local Cluster

Description

Run Malaria Simulation with Local Cluster

Usage

```
local_cluster_malariasim_controller(input, reps)
```

Arguments

`input`	Input parameters list
`reps`	Number of replicates

Value

List of simulation results

local_malariasim_controller

Run Malaria Simulation Locally

Description

Run Malaria Simulation Locally

Usage

```
local_malariasim_controller(input, reps)
```

Arguments

`input`	Input parameters list
`reps`	Number of replicates

Value

List of simulation results

resistance_to_dn0 *Convert resistance level to dn0 using spline fit*

Description

Convert resistance level to dn0 using spline fit

Usage

```
resistance_to_dn0(spline_fit, resistance_level)
```

Arguments

spline_fit	A smooth.spline object
resistance_level	Numeric resistance level to predict dn0 for

Value

Predicted dn0 value

resistance_to_overall_dn0 *Calculate weighted average dn0 across net types*

Description

Calculate weighted average dn0 across net types

Usage

```
resistance_to_overall_dn0(splines, usage_values, resistance_level)
```

Arguments

splines	Named list of smooth.spline objects for each net type
usage_values	Numeric vector of usage proportions for each net type
resistance_level	Numeric resistance level to calculate dn0 for

Value

Weighted average dn0 value

run_emulator — *Run Emulator for a Parameter*

Description

Run Emulator for a Parameter

Usage

```
run_emulator(
  db_path,
  param_index,
  models,
  window_size = 14,
  counterfactual = NULL,
  output_dir = NULL,
  plot_tight = FALSE,
  model_types = c("GRU", "LSTM")
)
```

Arguments

db_path	Path to DuckDB database
param_index	Parameter index to analyze
models	List from load_emulator_models()
window_size	Window size for rolling average
counterfactual	Named list with parameter name and values (optional)
output_dir	Directory to save outputs
plot_tight	Use tight y-axis scaling
model_types	Vector of model types to use

Value

List with predictions and plot

run_emulator_db — *Run Emulator Database Mode (Internal)*

Description

Run Emulator Database Mode (Internal)

Usage

```
run_emulator_db(
  db_path,
  param_index,
  models,
  window_size = 14,
  counterfactual = NULL,
  model_types = c("GRU", "LSTM")
)
```

Arguments

`db_path`	Path to DuckDB database
`param_index`	Parameter index to analyze
`models`	List from load_emulator_models()
`window_size`	Window size for rolling average
`counterfactual`	Named list with parameter name and values (optional)
`model_types`	Vector of model types to use

Value

List with predictions and metadata

`run_malariasim` *Run Malaria Simulations*

Description

Run malaria simulations for all parameter sets in the LHS scenarios file. This function handles dependency checking, parallel processing, and progress tracking.

Usage

```
run_malariasim(
  max_threads = 12,
  lhs_scenario = "Data/malariasim_scenarios.csv",
  bednet_params_path = NULL,
  output_dir = NULL,
  reps = 8,
  human_population = 1e+05,
  sim_years = 12
)
```

Arguments

`max_threads`	Maximum number of parallel workers to use
`lhs_scenario`	Path to LHS scenarios CSV file (default: "Data/malariasim_scenarios.csv")
`bednet_params_path`	Path to bednet parameters RDS file. If NULL (default), uses the bednet parameters bundled with the package.

`output_dir`	Directory to save simulation outputs (default: "Data")
`reps`	Number of replicates per parameter set (default: 8)
`human_population`	Human population size (default: 100000)
`sim_years`	Number of years to simulate (default: 12)

Value

A list with simulation results and metadata

`run_malaria_emulator` *Run Malaria Emulator*

Description

Runs malaria emulator in either database mode or scenario mode.

Usage

```
run_malaria_emulator(
  db_path = NULL,
  param_index = NULL,
  scenarios = NULL,
  predictor = "prevalence",
  models_base_dir = NULL,
  counterfactual = NULL,
  window_size = 14,
  device = NULL,
  model_types = c("GRU", "LSTM"),
  time_steps = 2190
)
```

Arguments

`db_path`	Path to DuckDB database (for database mode)
`param_index`	Parameter index for database mode (NULL for random selection)
`scenarios`	Data frame with scenario parameters (for scenario mode)
`predictor`	"prevalence" or "cases"
`models_base_dir`	Base directory with trained models. If NULL (default), uses models bundled with the package.
`counterfactual`	Named list for counterfactual analysis (database mode only)
`window_size`	Window size for rolling average
`device`	"cpu" or "cuda"
`model_types`	Vector of model types to use ("GRU", "LSTM", or both)
`time_steps`	Number of time steps for predictions (in days)

Value

Data frame with columns: index, timestep, value, model_type `run_minter_scenarios` *Run MINT Scenarios for Malaria Intervention Analysis*

Description

Run MINT Scenarios for Malaria Intervention Analysis

Usage

```
run_minter_scenarios(
  res_use,
  res_future = NULL,
  py_only,
  py_pbo,
  py_pyrrole,
  py_ppf,
  net_type_future = NULL,
  itn_future = NULL,
  prev,
  Q0,
  phi,
  season,
  routine,
  irs,
  irs_future,
  lsm,
  eir_models = "xgboost",
  prevalence_models = "LSTM",
  predictor = c("prevalence", "cases"),
  year_start = 2,
  year_end = 5,
  scenario_tag = NULL
)
```

Arguments

`res_use`	Numeric vector of current resistance levels
`res_future`	Numeric vector of future resistance levels after next campaign
`py_only`	Numeric vector of pyrethroid-only net proportions
`py_pbo`	Numeric vector of pyrethroid-PBO net proportions
`py_pyrrole`	Numeric vector of pyrethroid-pyrrole net proportions
`py_ppf`	Numeric vector of pyrethroid-PPF net proportions
`net_type_future`	Character vector of future net choices per scenario (allowed: "py_only","py_pbo","py_pyrrole","py_p use NA for baseline.
`itn_future`	Numeric vector (0–1) of future ITN coverage per scenario; use NA for baseline.
`prev`	Numeric vector of malaria prevalence levels
`Q0`	Numeric vector of Q0 values

`phi`	Numeric vector of phi bednet values
`season`	Numeric vector of seasonality indicators
`routine`	Numeric vector of routine treatment indicators
`irs`	Numeric vector of current IRS coverage
`irs_future`	Numeric vector of future IRS coverage
`lsm`	Numeric vector of LSM coverage
`eir_models`	Character vector of models to use for EIR prediction (default: "xgboost")
`prevalence_models`	Character vector of models for prevalence prediction (default: "LSTM")
`predictor`	Character vector of predictors to run (default: c("prevalence", "cases"))
`year_start, year_end`	Integers for case-year range
`scenario_tag`	Optional character vector of scenario identifiers. If provided, it must have length equal to the number of scenarios. Defaults to "Scenario1", "Scenario2", ...

Value

List of prevalence and cases predictions

---

run_mintweb_scenarios *Run MINT Scenarios for Malaria Intervention Analysis*

Description

Run MINT Scenarios for Malaria Intervention Analysis

Usage

```
run_mintweb_scenarios(
  res_use,
  res_future = NULL,
  py_only,
  py_pbo,
  py_pyrrole,
  py_ppf,
  net_type_future = NULL,
  itn_future = NULL,
  prev,
  Q0,
  phi,
  season,
  routine,
  irs,
  irs_future,
  lsm,
  eir_models = "xgboost",
  prevalence_models = "LSTM",
  predictor = c("prevalence", "cases"),
```

```
  year_start = 2,
  year_end = 5,
  scenario_tag = NULL,
  cull_prevalence = c(2, 7),
  output_time = FALSE
)
```

Arguments

- `res_use` Numeric vector of current resistance levels
- `res_future` Numeric vector of future resistance levels after next campaign
- `py_only` Numeric vector of pyrethroid-only net proportions
- `py_pbo` Numeric vector of pyrethroid-PBO net proportions
- `py_pyrrole` Numeric vector of pyrethroid-pyrrole net proportions
- `py_ppf` Numeric vector of pyrethroid-PPF net proportions
- `net_type_future` Character vector of future net choices per scenario (allowed: "py_only","py_pbo","py_pyrrole","py_p use NA for baseline.
- `itn_future` Numeric vector (0–1) of future ITN coverage per scenario; use NA for baseline.
- `prev` Numeric vector of malaria prevalence levels
- `Q0` Numeric vector of Q0 values
- `phi` Numeric vector of phi bednet values
- `season` Numeric vector of seasonality indicators
- `routine` Numeric vector of routine treatment indicators
- `irs` Numeric vector of current IRS coverage
- `irs_future` Numeric vector of future IRS coverage
- `lsm` Numeric vector of LSM coverage
- `eir_models` Character vector of models to use for EIR prediction (default: "xgboost")
- `prevalence_models` Character vector of models for prevalence prediction (default: "LSTM")
- `predictor` Character vector of predictors to run (default: c("prevalence", "cases"))
- `year_start, year_end` Integers for case-year range
- `scenario_tag` Optional character vector of scenario identifiers. If provided, it must have length equal to the number of scenarios. Defaults to "Scenario1", "Scenario2", ...
- `cull_prevalence` Optional integer vector to remove the first X years worth of simulation and aggregate
- `output_time` Optional bool to cull absolute time tracking from output objects

Value

List of prevalence and cases predictions

set_bednet_parameters *Set Bednet Parameters*

Description

Set Bednet Parameters

Usage

```
set_bednet_parameters(simparams, lhs_sample, bednet_params, baseline = TRUE)
```

Arguments

simparams	Simulation parameters object
lhs_sample	Single row from LHS data
bednet_params	Bednet parameter data
baseline	Logical, whether this is baseline (default TRUE)

Value

List with updated simparams and timesteps

set_irs_parameters *Set IRS Parameters*

Description

Set IRS Parameters

Usage

```
set_irs_parameters(simparams, lhs_sample)
```

Arguments

simparams	Simulation parameters object
lhs_sample	Single row from LHS data

Value

List with updated simparams and timesteps

set_lsm_parameters *Set LSM Parameters*

Description

Set LSM Parameters

Usage

```
set_lsm_parameters(simparams, lhs_sample)
```

Arguments

simparams	Simulation parameters object
lhs_sample	Single row from LHS data

Value

List with updated simparams and timesteps

set_mosquito_parameters *Set Mosquito Parameters*

Description

Set Mosquito Parameters

Usage

```
set_mosquito_parameters(lhs_sample)
```

Arguments

lhs_sample	Single row from LHS data

Value

List of mosquito parameters

`set_seasonality` *Set Seasonality Parameters*

Description

Set Seasonality Parameters

Usage

```
set_seasonality(lhs_sample)
```

Arguments

`lhs_sample`	Single row from LHS data

Value

List with g0, g, and h parameters

Index